

Yusuf Kayra Gül

Undergraduate Physics Student

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Education

- 2023–Present **B.S. in Physics**, *Bilkent University*, Ankara, Turkey, High Honour Student (5 Semesters)
- 2019–2023 **High School Diploma**, *Sümer Science High School (Sümer Fen Lisesi)*, Kayseri, Turkey

Scholarships & Awards

- 2023–Present **Bilkent University Comprehensive Merit Scholarship**, *Bilkent University*, Ankara, Turkey
- Awarded upon university admission for ranking 3,369th among roughly 3 million candidates in Türkiye's National University Entrance Exam (YKS).
- 2023–Present **TÜBİTAK 2205 Undergraduate Scholarship**, *TÜBİTAK*, Ankara, Turkey
- Awarded by TÜBİTAK for the mentioned achievement in YYS.
- 2024–Present **TOG Bezmen Science Scholarship**, *TOG (Toplum Gönüllüleri Vakfı – Community Volunteers Foundation)*, Turkey
- Granted for maintaining GPA above 3.50 in Physics.
- 2026 Summer **TÜBİTAK 2224-D Grant for Participation in Scientific Education Activities Abroad**, *TÜBİTAK*
- Granted for pursuing an internship on Cavity Quantum Electrodynamics at Universität Innsbruck.

Projects

- 2020 & 2021 **Teknofest UAV Competition (Team SÜMERİHA)**, *İnter High School, Rotary Wing Category Finalist*, Turkey
- Served as the team pilot and collaborated on the design, manufacturing, and testing of a rotary-wing UAV. Designed UAV chassis using SOLIDWORKS.
 - Achieved finalist status for two consecutive years (2020 and 2021) in the Rotary Wing Category.
- 2024–2025 Spring **Observation of Black-body Radiation and Flame Emission Spectra of Alkali Metals by Using Visible Light Spectroscopy**, *Modern Physics Course Project*
- Successfully constructed a home-made visible light spectrometer to measure black-body radiation of a tungsten filament and flame emission spectra of Li, K, and Na with ± 5 nm precision.
 - [View Report on GitHub](#) | [Watch Demonstration on YouTube](#)

- 2025–2026 **Vibration Analysis of a Loudspeaker via Michelson Interferometer**, *Waves and Optics Course Project*, **Best Project Award Winner**
- Built a modified Michelson interferometer with a loudspeaker-mounted mirror to measure vibration frequencies through interference pattern modulation.
 - Acquired high rate photodetector data using Arduino and extracted vibration frequencies via FFT-based signal processing.
 - [View Report on GitHub](#) | [Watch Demonstration on YouTube](#)
- 2025–2026 **Quantum Teleportation State Explorer**, *Quantum Optics Course Project*
- Spring
- Developed a Python-based graphical application and standalone browser demo for studying state optimization in continuous-variable quantum teleportation.
 - [View Report on GitHub](#)

Experience

- 2025 **Peer Tutor (PHYS101 – General Physics I)**, *Bilkent University Teaching & Learning Support Center (BTLSC)*, Ankara, Turkey
- Provided approximately 20 hours of peer tutoring for undergraduate students in PHYS101 (General Physics I).
 - Completed certified Peer Tutor Training (4 hours) covering effective teaching strategies and student-centered learning.
- 2026 Summer **Research Intern**, *Institut für Theoretische Physik, Universität Innsbruck*, Innsbruck, Austria
- Conducted research in the Cavity Quantum Electrodynamics (CQED) group under the supervision of Prof. Helmut Ritsch, Dr. Farokh Mivehvar, and Dr. Natalia Masaleva.
 - Studied the steady state dynamics of a particle coupled to an optical cavity under artificial Gauge field.

Extracurricular Activities

- 2025–Present **Board Member**, *Bilkent Physics Society*, Bilkent University
- Co-organized academic seminars, social events, and outreach programs for physics students.
- 2025–Present **Editor**, *Bilkent Science Literacy Society*, Bilkent University
- Currently reviewing and fact-checking senior-level articles about physics.

Skills

- Languages Turkish (Native), English (Fluent – C1)
- Programming Python (NumPy, SciPy, QuTiP, Matplotlib), Julia (QuantumOptics.jl, StochasticDifferentialEq.jl), MATLAB, Arduino
- Software MS Office Suite, LaTeX, SOLIDWORKS (Beginner), Theremino Spectrometer, Tracker Video Analysis and Modeling Tool
- Licenses & Certifications **Type IHA-1 Sportif/Amateur Drone Pilot License** – Directorate General of Civil Aviation (SHGM).
- PCAP: Programming Essentials in Python** – Cisco Networking Academy® & MoNE, June 2020.
- Akıllı Tekno Digital Competence Certificate** – Ministry of Industry and Technology, issued March 4, 2024. [View Credential](#)

Innovation and Entrepreneurship Competence Certificate – Ministry of Industry and Technology, issued April 3, 2024. *View Credential*

YGA Zirvesi 2025 Participation Certificate – Young Guru Academy (YGA), December 2025. *View Credential*